

Companion essay

Arithmetic Inherits the Geometry of its Basis

Euler, Pick, determinants, and the passage from Gaussian to Eisenstein integers

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Interactive companion: leelemacjames.github.io/euler_pick_lattice_learner_v3.html

Abstract

The four ideas of Euler's topological invariant, Pick's lattice formula, the determinant as area, and the passage from the square lattice to the triangular lattice of Eisenstein are usually separate notions. They are facets of one thing, and this essay traces the single thread that connects them: the determinant of a lattice basis is the fundamental unit of area in that lattice, and every geometric formula expressed in that lattice inherits the determinant as its scaling factor. The square lattice is not the default. It is the exceptional case in which the determinant happens to equal one.

How Euler's formula survives deformation

Take any convex polyhedron such as a cube, a tetrahedron, a dodecahedron. Count its vertices V , edges E , and faces F . No matter what shape you choose,

$$V - E + F = 2.$$

Rescale the polyhedron, stretch it, or shear it into almost any form you please, and the relation remains untouched. The lengths of edges and the angles between faces may change dramatically, yet the underlying pattern of connections survives. Vertices remain attached to the same edges, edges continue to separate the same faces, and that persistent network of adjacencies is exactly what topology seeks to capture. The constant 2 is the Euler characteristic χ of the sphere, and every convex polyhedron inherits it because each may be deformed continuously into a sphere without tearing or puncturing its surface.

Topology preserves adjacency, not shape. That is why Euler's constant survives deformation.

The proof makes this concrete. Project the polyhedron onto a plane and imagine shining a light through a transparent model and reading off the shadow. The three-dimensional solid becomes a planar graph, and the same count $V - E + F = 2$ holds for any planar graph. The dimension changed; the invariant did not.

This planar projection is the first hint of a pattern: the Euler formula will reappear, doing real work, when we turn to a completely different-looking problem about counting lattice points inside polygons.

Counting points to measure area

Draw a polygon on squared paper with all corners at grid intersections. Let I be the number of grid points strictly inside it and B the number lying exactly on the boundary. Pick's theorem says the area is

$$A = I + \frac{B}{2} - 1.$$

The formula looks miraculous. Area as a continuous, geometric quantity that emerges from a count of discrete points. The miracle dissolves when you see the proof, which turns out to be an application of the same Euler formula we just met.

The proof strategy

Subdivide the polygon into *primitive triangles*: triangles whose only lattice points are the three vertices themselves. Apply Euler's formula to the resulting planar graph. Let T be the number of triangles. With $V = I + B$ total lattice points and faces $F = T + 1$ (the triangles plus the single unbounded outer face), Euler gives $V - E + (T + 1) = 2$, so

$$T = E - V + 1.$$

Count edges two ways. Each interior edge is shared by two triangles; each boundary edge borders only one. This gives $3T = 2E - B$, so $E = (3T + B)/2$. Substituting back:

$$T = \frac{3T + B}{2} - (I + B) + 1 \implies T = 2I + B - 2.$$

The area formula now hinges on a single geometric fact about each primitive triangle.

Why every primitive triangle has area exactly $\frac{1}{2}$

A triangle has area $\frac{1}{2} \times \text{base} \times \text{height}$. Place one vertex at the origin and write the other two as vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. The base is $|\mathbf{v}_1|$ and the height is $|\mathbf{v}_2| \sin \theta$, where θ is the angle between them:

$$A_{\triangle} = \frac{1}{2} |\mathbf{v}_1| |\mathbf{v}_2| \sin \theta = \frac{1}{2} |\det[\mathbf{v}_1, \mathbf{v}_2]| = \frac{1}{2} \left| \det \begin{bmatrix} v_{1x} & v_{2x} \\ v_{1y} & v_{2y} \end{bmatrix} \right|.$$

The determinant is both an algebraic device and the *signed area* of the parallelogram spanned by the two vectors. The formula $\frac{1}{2} |\det|$ is just base times height in coordinate-free language.

A triangle is *primitive* precisely when its edge vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ form a *basis* for $\mathbb{Z}^2$. This means every lattice point can be written as an integer combination of them. This being exactly the condition $|\det[\mathbf{v}_1, \mathbf{v}_2]| = 1$. Such a matrix is called **unimodular**.

The pivotal fact: every primitive triangle has area $\frac{1}{2}$ because the unimodular condition forces $|\det| = 1$, making the triangle exactly half a unit parallelogram.

Since $A = T/2$ and $T = 2I + B - 2$:

$$A = I + \frac{B}{2} - 1. \quad \square$$

So we arrive at the conclusion that Pick's theorem is essentially a theorem about determinants.

*A basis chooses directions. The determinant measures the area enclosed by those directions.
Every lattice formula inherits that choice of area.*

The determinant as a geometric measure

Let's step back and press on what this determinant construct actually *means*.

Given a 2×2 matrix

$$M = \begin{bmatrix} a & c \\ b & d \end{bmatrix},$$

the determinant $\det(M) = ad - bc$ measures *oriented area scaling*: if you apply M to the unit square, the image has area $|\det(M)|$. For a basis matrix whose columns are the two vectors that generate a lattice, the determinant is the area of the *fundamental parallelogram*, the unit cell that tiles the plane.

A basis is a ruler

Any lattice begins with a choice of two generating directions. Every point is then reached by taking integer combinations of those vectors. The arithmetic remains the same whichever basis is chosen, but the geometry through which that arithmetic is viewed need not. A different basis carries with it a different fundamental cell, and therefore a different notion of unit area.

The area of the fundamental cell depends on exactly three quantities: the length of the first basis vector, the length of the second, and the angle between them:

$$\Delta = |\mathbf{b}_1| |\mathbf{b}_2| \sin \theta.$$

The determinant is the angle-corrected product of lengths and therefore depends only on two lengths and the angle between them. When the basis vectors happen to meet at right angles, the sine term disappears. When they also happen to have unit length, the determinant reduces to one. The square lattice occupies this particularly simple corner of a much larger family. The hierarchy of special cases is:

Basis type	Lengths	Angle	Δ
General	arbitrary	arbitrary	$ \mathbf{b}_1 \mathbf{b}_2 \sin \theta$
Orthogonal	arbitrary	90°	$ \mathbf{b}_1 \mathbf{b}_2 $
Orthonormal	unit	90°	1
Eisenstein	unit	60°	$\sqrt{3}/2$

The square lattice is thus (just) the orthonormal special case.

Why the determinant disappears on the square lattice

On the integer square lattice $\mathbb{Z}^2$:

$$M_{\square} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \det(M_{\square}) = 1.$$

The classical Pick formula $A = I + B/2 - 1$ carries no prefactor. But the determinant has not vanished. It has become the unit by which all other areas are measured. The determinant is hidden in the same way that multiplying by one is hidden.

In the Gaussian integers this geometric factor passes almost unnoticed, because the standard basis is orthonormal and the determinant equals one. Rotating one basis vector through sixty degrees brings it back into view. Orthonormality is merely the exceptional case in which the determinant happens to equal one.

A second lattice: the Eisenstein integers

Nothing in arithmetic requires the basis vectors to meet at right angles. The familiar square grid owes more to the convenience of the page than to any intrinsic property of arithmetic.

Rotate one basis vector through 60° while keeping its length unchanged. Every integer combination still lands on a lattice point and the arithmetic survives while the geometry changes.

The *Gaussian integers* $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ place basis vectors at 90° , reproducing the square lattice. Now let

$$\omega = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

be a primitive cube root of unity, satisfying $\omega^3 = 1$ and $\omega^2 + \omega + 1 = 0$.

The *Eisenstein integers*

$$\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$$

form a triangular lattice which is the densest packing of equal circles in the plane with basis vectors 1 and ω meeting at 60° . The basis matrix is

$$M_\omega = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} \end{bmatrix}, \quad \det(M_\omega) = 1 \cdot \frac{\sqrt{3}}{2} - 0 \cdot \left(-\frac{1}{2}\right) = \frac{\sqrt{3}}{2}.$$

The fundamental rhombus with angles 60° and 120° has area $\sqrt{3}/2$, not 1 and its lattice is *not* unimodular.¹

Comparing the two basis matrices:

Square lattice $\mathbb{Z}^2$

$$M_\square = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\det = 1 \times 1 - 0 \times 0 = 1$$

Basis at 90° . Unimodular.

Det hidden: equals the unit of measure.

$$\text{Primitive triangle area} = \frac{1}{2}.$$

Eisenstein lattice $\mathbb{Z}[\omega]$

$$M_\omega = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$\det = 1 \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

Basis at 120° . Not unimodular.

Det revealed: differs from unity.

$$\text{Primitive triangle area} = \frac{\sqrt{3}}{4}.$$

Interactive companion learner

(drag polygons, shear the lattice, watch the determinant and Pick formula update live):

leelemacjames.github.io/euler_pick_lattice_learner_v3.html

¹in the context of the Eisenstein lattice *itself*, the transition matrix between any two valid Eisenstein bases will have a determinant of ± 1 (or a unit in $\mathbb{Z}[\omega]$).

Pick's theorem on the Eisenstein lattice

Nothing essential in Pick's proof depends on the square lattice itself. The triangulation, the Euler count, and the edge-counting argument are entirely combinatorial. They survive the passage to the Eisenstein lattice without alteration. The only geometric ingredient is the area of a primitive triangle, and that quantity is fixed by the determinant of the chosen basis. The *only* line that changes is the area per primitive triangle.

Lattice	Primitive triangle area	Pick formula
Square ($\Delta = 1$)	$\frac{1}{2} \cdot 1 = \frac{1}{2}$	$A = I + \frac{B}{2} - 1$
Eisenstein ($\Delta = \frac{\sqrt{3}}{2}$)	$\frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$	$A = \frac{\sqrt{3}}{2} \left(I + \frac{B}{2} - 1 \right)$

The combinatorial expression $I + B/2 - 1$ is universal. The prefactor $\Delta = \det(M)$ encodes which lattice you are working in. For any lattice:

$$A = \Delta \cdot \left(I + \frac{B}{2} - 1 \right) \quad \Delta = \det(M_{\text{basis}}).$$

The classical Pick formula is the special case $\Delta = 1$.

Pick's theorem is one formula, not two. The lattice determinant Δ is the single parameter distinguishing them. Arithmetic inherits the geometry of its basis.

The cosine rule confirms the determinant

There is a satisfying elementary check we can perform. The determinant of the Eisenstein basis matrix equals $|\mathbf{f}_1||\mathbf{f}_2|\sin\theta = 1 \cdot 1 \cdot \sin 60^\circ = \sqrt{3}/2$. That is the cross-product area formula. But the same answer emerges from the cosine rule. The diagonal of the fundamental rhombus has length

$$d^2 = |\mathbf{f}_1|^2 + |\mathbf{f}_2|^2 - 2|\mathbf{f}_1||\mathbf{f}_2|\cos 60^\circ = 1 + 1 - 1 = 1,$$

so $d = 1$. The rhombus splits into two isosceles triangles each of area $\frac{1}{2} \sin 60^\circ = \frac{\sqrt{3}}{4}$, giving rhombus area $\frac{\sqrt{3}}{2}$.

Thus, both the determinant and the cosine rule arrive at the same value, measuring the same fundamental cell from different directions.

Retracing our thread

The ideas here may seem to belong to different parts of mathematics as Euler's formula feels like topology and Pick's theorem feels like combinatorial geometry. Determinants feel like linear algebra while Gaussian and Eisenstein integers feel like number theory. We now see them as not so separate. Euler's formula supplies the combinatorial scaffold. Pick's theorem places that scaffold on a lattice. The determinant then enters quietly, fixing the area carried by each primitive triangle. Once that role is recognised, the passage from the square lattice to the Eisenstein lattice requires changing only a single number. The passage from $\mathbb{Z}^2$ to $\mathbb{Z}[\omega]$ makes this concrete. The 90° basis of the Gaussian integers gives $\det = 1$. The 60° basis of the Eisenstein integers gives $\det = \sqrt{3}/2$. The Pick formula scales by that factor.

$$A = \Delta \cdot \left(I + \frac{B}{2} - 1 \right)$$

where $\Delta = \det(M_{\text{basis}})$ is the area of the fundamental domain.

This is what it means to say that arithmetic inherits the geometry of its basis. Changing the basis changes the fundamental unit of area, and every lattice formula inherits that choice through the determinant. We will stop before wading into deeper waters, but it is worth gesturing toward the depths.

In the Gaussian and Eisenstein cases, the determinant is more than an isolated geometric quantity being the Euclidean shadow of a deeper arithmetic invariant: the *field discriminant*. The geometry of the lattice and the factorisation properties of the ring are two descriptions of the same underlying structure. The Eisenstein integers $\mathbb{Z}[\omega]$ form a unique factorisation domain. The class number of the number field $\mathbb{Q}(\omega)$ is 1, which is the algebraic statement that unique factorisation holds. The lattice determinant $\sqrt{3}/2$ is the Euclidean trace of that arithmetic fact being the discriminant of the field, rendered as the area of a rhombus. The passage from Gaussian to Eisenstein integers as a change of grid, reflects a much broader principle: arithmetic carries its own geometry. The Gaussian integers $\mathbb{Z}[i]$ have the same property. As do the rings associated with several other quadratic fields. The lattices for which this works most cleanly are connected to the Heegner numbers, which is an often told [story](#) linking unique factorisation, class numbers, and the remarkable near-integers produced by expressions such as $e^{\pi\sqrt{163}}$.

The thread that began by counting the corners of a polyhedron eventually arrives at some of the deepest structures in number theory. Along the way the determinant never changes its role. It measures the fundamental cell of the chosen basis, whether that basis belongs to the square lattice, the Eisenstein lattice, or the higher-dimensional lattices arising from algebraic number fields. The familiar Cartesian grid simply makes that geometry easy to overlook.